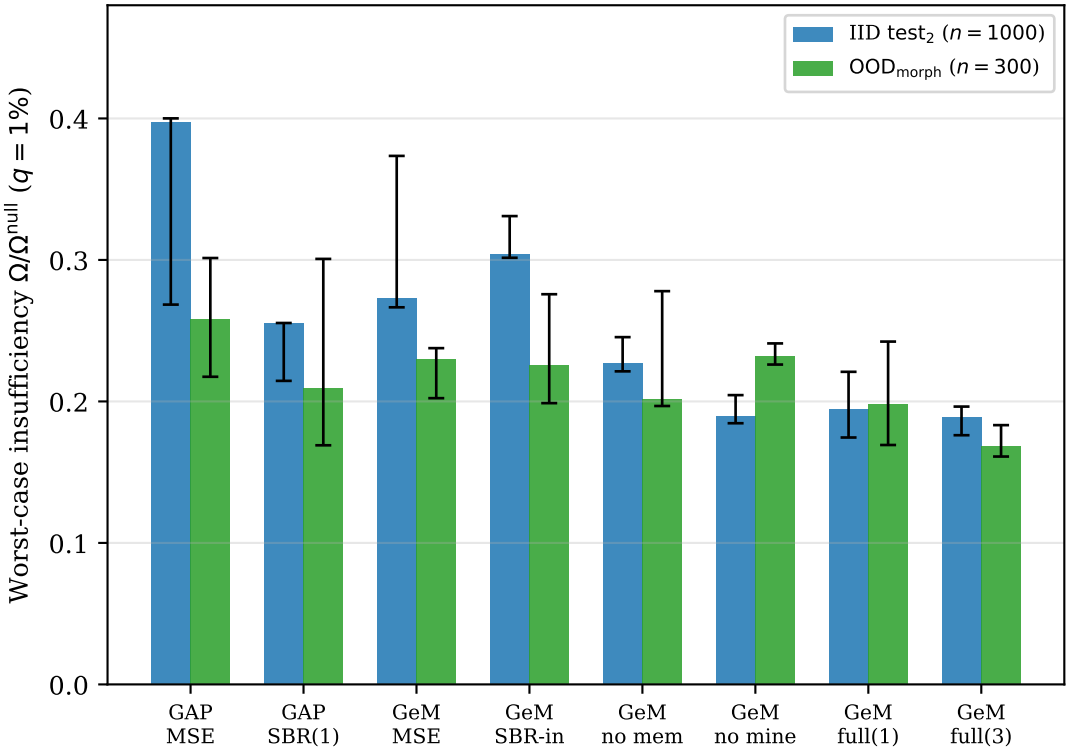
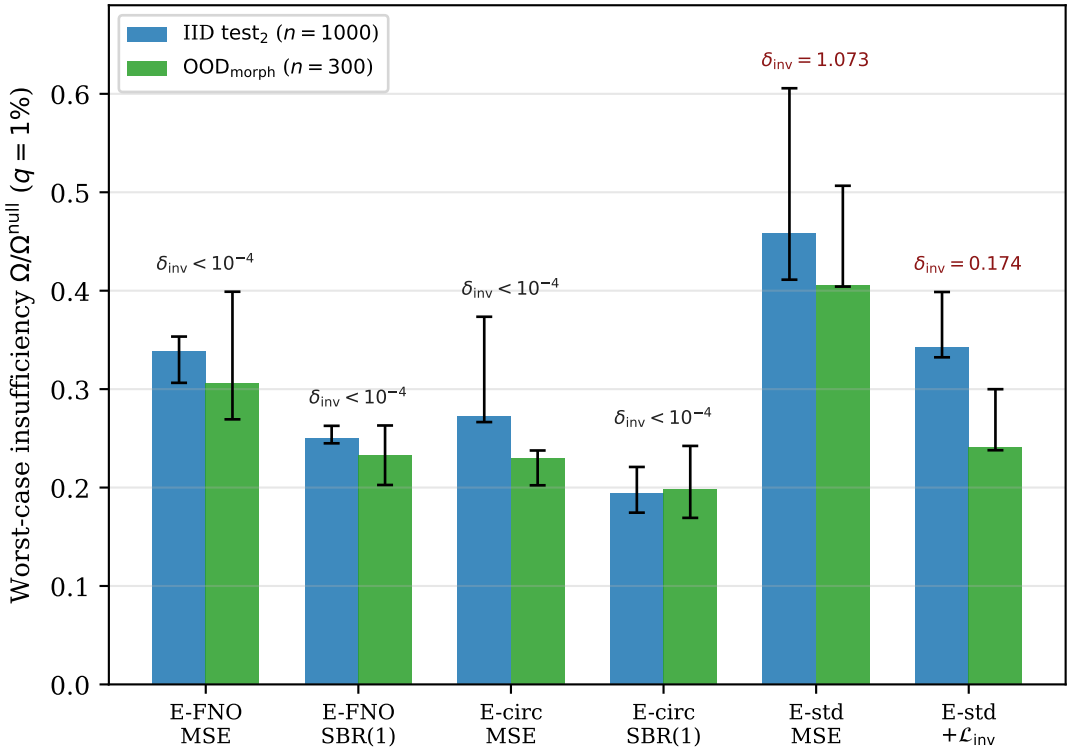

(a) Objective & pooling ablations on E-circ ($d = 8$)
IID test$_2$ ($n = 1000$)
OOD$_{\text{morph}}$ ($n = 300$)
Worst-case insufficiency $\Omega/\Omega^{\text{null}}$ ($q = 1\%$)
0.0
0.1
0.2
0.3
0.4
GAP MSE
GAP SBR(1)
GeM MSE
GeM SBR-in
GeM no mem
GeM no mine
GeM full(1)
GeM full(3)


(b) Backbone transfer (E-FNO) & invariance (E-std)
IID test$_2$ ($n = 1000$)
OOD$_{\text{morph}}$ ($n = 300$)
Worst-case insufficiency $\Omega/\Omega^{\text{null}}$ ($q = 1\%$)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
$\delta_{\text{inv}} < 10^{-4}$
$\delta_{\text{inv}} < 10^{-4}$
$\delta_{\text{inv}} < 10^{-4}$
$\delta_{\text{inv}} < 10^{-4}$
$\delta_{\text{inv}} = 1.073$
$\delta_{\text{inv}} = 0.174$
E-FNO MSE
E-FNO SBR(1)
E-circ MSE
E-circ SBR(1)
E-std MSE
E-std $+\mathcal{L}_{\text{inv}}$